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ORIGINAL PAPER

Solving methods for interval linear programming problem: a review and an improved method

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Abstract Interval linear programming is used for tackling interval uncertainties in real-world systems. An arbitrary point is a feasible point to the interval linear programming model if it lies in the largest feasible region of the interval linear programming model, and it is optimal if it is an optimal solution to a characteristic model. The optimal solution set to the interval linear programming is the union of all solutions that are optimal for a characteristic model. In this paper, we review some existing methods for solving interval linear programming problems. Using these methods the interval linear programming model is transformed into two sub-models. The optimal solutions of these sub-models form the solution space of these solving methods. A part of the solution space of some of these methods may be infeasible. To eliminate the infeasible part of the solution space of above methods, several methods have been proposed. The solution space of these modified methods may contain non-optimal solutions. Two improvement methods have been proposed to remove the non-optimal solutions of the solution space of above modified methods. Finally, we introduce an improved method and its sub-models. The solution space of our method is absolutely both feasible and optimal.

Keywords Feasibility · Optimality · Interval linear programming · Stability · Optimal solution set

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1 Introduction

In the real world, system parameters may be uncertain. Interval linear programming is an efficient device for decision making under uncertainty. Recently, the majority of researches have been centralized on solving interval linear programming (ILP) models (Allahdadi and Mishmast Nehi 2013, 2015; Allahdadi et al. 2016; Ashayerinasab et al. 2018; Beeck 1978; Chinneck and Ramadan 2000; Fan and Huang 2012; Hladik 2014; Huang et al. 1995; Huang and Moore 1993; Huang and Cao 2011; Jansson and Rump 1991; Konickova 2001; Lu et al. 2014; Mraz 1996; Tong 1994; Wang and Huang 2014; Zhou et al. 2009). Some of these works used interval arithmetic (Beeck 1978; Jansson 1988; Jansson and Rump 1991; Machost 1970), whereas others have focused on B-stability (basis stability) (Hladik 2014; Konickova 2001; Mraz 1996; Rohn 1993b). A given point is a feasible solution to the ILP model, if it satisfies the best model constraints (i.e., $A^{-}\mathbf{x} \leq \mathbf{b}^{+}, \mathbf{x} \geq 0$) and it is an optimal solution to the ILP model, if it is an optimal solution to at least an arbitrary characteristic model of the ILP model. By assumption of B-stability, the optimal solution set to the ILP model is determined by the region created by the best model constraints and the worst model constraints with inverse signs, when the feasible solution components of the best model are positive (Allahdadi and Mishmast Nehi 2013). In Hladik (2014), the authors proved that under assumption of unique B-stability, the exact optimal solution set to the ILP model is equal to the solution set to the interval system $\mathbf{A}_B^{-}\mathbf{x}_B \leq \mathbf{b}^{+}, \mathbf{A}_B^{+}\mathbf{x}_B \geq \mathbf{b}^{-}, \mathbf{x}_B \geq \mathbf{0}, \mathbf{x}_N = \mathbf{0}$.

Some of the solving methods transform the ILP model into two sub-models whereas their optimal solutions form a set which is called solution space of the ILP model (Fan and Huang 2012; Huang et al. 1995; Huang and Moore 1993; Lu et al. 2014; Tong 1994; Wang and Huang 2014; Zhou et al. 2009). In the best and worst cases (BWC) method presented by Tong (1994), the ILP model was transformed into two sub-models, the best and worst sub-models, which consist of the largest and smallest feasible regions, respectively. The best and worst objective function values can be obtained by the BWC method. The BWC method introduces exact bounds of the optimum objective function values. However, a part of solution space of the BWC method may be infeasible. Chinnect and Ramadan developed BWC method when ILP model includes equality constraints (Chinneck and Ramadan 2000). Grey system theory proposes a method for insert uncertainty into systems analysis. A novel method of solution for gray linear programming was proposed by Huang and Moore (1993). This method provides the solutions of interval objective function and decision variables.

Among methods for solving ILP model, two-step method (TSM) have been presented by Huang et al. (1995). In recent years, hundreds of research papers have been published based on the TSM method to use interval uncertainties in real world problems (Cai et al. 2007; Fan and Huang 2012; Guo and Huang 2009a, b; He et al. 2009; Huang 1996; Huang et al. 1994; Huang and Cao 2011; Lu et al. 2008; Luo et al. 2005; Zhou et al. 2009). The solution space of the TSM method may include infeasible solutions. To eliminate infeasible solutions from the solution space of the TSM method, some methods have been proposed. Zhou et al. (2009) exhibited modified interval linear programming (MILP) method, adding an extra constraint to the second sub-model of the TSM method. The solution space of the MILP method is absolutely feasible, but some of the solutions obtained using the MILP method may be non-optimal. Huang and Cao (2011) presented three-step method (ThSMs) followed by a feasibility test to help the solution violation. If the solution space of the TSM method has not been absolutely feasible, a shrinking process for the solution space of the TSM method into centre point would be used to eliminate the infeasible solutions. These models are called ThSMs (ThSM-I and ThSM-II). The ThSMs models will be infeasible when the centre point of the solution space of the TSM method is infeasible.

Fan and Huang (2012) presented a robust two-step method (RTSM), using extra constraints, to remove the infeasible solutions from the solution space of the TSM method. Wang and Huang (2014) proposed an improved TSM method (ITSM), in which extra constraints are added to the second sub-model of the TSM method to ensure that the given solutions are absolutely feasible. Although, the solution spaces of the ITSM and RTSM methods are absolutely feasible adding extra constraints to the second sub-model, part of solution space of these methods is not optimal. Lu et al. (2014) proposed an alternative solution method (i.e., SOM-2) which is similar to the TSM method, generalized regarding different forms of interval parameters in the ILP model.

In this paper, we study about feasibility and optimality of solutions obtained using the solving methods. Hence, we introduce essential theorems for basis stability and exact optimal solution set to the ILP model. Then, we review some of the previous solving methods presented above and check both absolute feasibility and optimality of the solution spaces of those methods. After that, we introduce a method to improve SOM-2 method namely ISOM-2 method. Because the solution space of the SOM-2 method is not absolutely feasible and optimal in general. The solution space of the ISOM-2 method is both feasible and optimal. Finally, we solve a numerical example using above solving methods and compare their solution spaces with the exact optimal solution set to the ILP model.

Briefly, by comparing the solution space of the solving methods with exact optimal solution set to the ILP model, we obtain: (1) The solution spaces of the BWC, TSM, and SOM-2 methods may contain infeasible and/or non-optimal solutions. (2) To improve the solution space of the TSM method, The MILP, ITSM, RTSM, and ITSM methods are introduced. The solution spaces of these methods are absolutely feasible but are not absolutely optimal. (3) To obtain the solution space which is both absolutely feasible and optimal, the IMILP and IILP methods are introduced in Allahdadi et al. (2016), and the ISOM-2 method is introduced in this paper.

An interval number $x^{\pm}$ is generally denoted by $[x^-, x^+]$ where $x^- \leq x^+$. If $x^- = x^+$, then $x^{\pm}$ will degenerate. An interval matrix is defined as $\mathbf{A}^{\pm} = [A^-, A^+] = \{A \mid A^- \leq A \leq A^+\}$ where A^- and A^+ are two matrices in $\mathbb{R}^{m \times n}$ and $A^- \leq A^+$. We denote the centre and the radius of $\mathbf{A}$ with $\mathbf{A}^c = \frac{1}{2}(A^- + A^+)$ and $\Delta_{\mathbf{A}^{\pm}} = \frac{1}{2}(A^+ - A^-)$, respectively. The set of all interval matrices $(m \times n)$ is denoted by $\mathbb{IR}^{m \times n}$. A square matrix $\mathbf{A}^{\pm}$ is called regular if each $A \in \mathbf{A}^{\pm}$ is nonsingular. An interval vector $\mathbf{x}^{\pm} = \{\mathbf{x} \mid \mathbf{x}^{-} \leq \mathbf{x} \leq \mathbf{x}^{+}\}$, where $\mathbf{x}^{-}, \mathbf{x}^{+} \in \mathbb{R}^{n}$ is an one column interval matrix.

2 Preliminaries

In this section, we introduce general form of the ILP model, some essential theorems for basis stability and exact optimal solution set to the ILP model.

2.1 General form of ILP

A general form of ILP model is considered as follows:

$$\begin{aligned} \max f^{\pm} &= \sum_{j=1}^{n} c_j^{\pm} x_j^{\pm} \\ \text{s. t.} \quad & \sum_{j=1}^{n} a_{ij}^{\pm} x_j^{\pm} \leq b_i^{\pm}, \quad i = 1, \ldots, m, \\ & x_j^{\pm} \geq 0, \quad j = 1, \ldots, n. \end{aligned} \tag{1}$$

An arbitrary characteristic model of ILP model (1) can be expressed as follows:

$$\begin{aligned} \max f = & \sum_{j=1}^{n} c_j x_j \\ \text{s. t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i = 1, \ldots, m, \\ & x_j \geq 0, \quad j = 1, \ldots, n. \end{aligned} \tag{2}$$

where $c_j \in c_j^{\pm} = [c_j^{-}, c_j^{+}]$, $a_{ij} \in a_{ij}^{\pm} = [a_{ij}^{-}, a_{ij}^{+}]$, and $b_i \in b_i^{\pm} = [b_i^{-}, b_i^{+}]$.

Definition 1 The point $\mathbf{x}^0 = (x_1^0, x_2^0, \ldots, x_n^0) \in \mathbb{R}^n$ is a feasible solution to the ILP model (1), if it satisfies in the best model constraints which have the largest feasible region(i.e., $\mathbf{x}^0 \in Q = \{\mathbf{x} \mid A^{-}\mathbf{x} \leq \mathbf{b}^{+}, \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \geq 0\}$, where Q has been defined as the largest feasible solution region of the ILP model (1).

Definition 2 A zone $\hat{X}$ is an extreme zone of the feasible region of the ILP model if each of the its points is an extreme point of the feasible region of a given characteristic model of the ILP model.

Definition 3 A point $\bar{\mathbf{x}}$ is an optimal solution to the ILP model if it is an optimal solution to a given characteristic model of the ILP model. The exact optimal solution set to the ILP model is the union of all solutions that are optimal solution to a given characteristic model.

2.2 Basis stability and optimal solution set to the ILP model

In this subsection, we define basis stability and its conditions. Under these conditions, the optimal solution set to the ILP model will be determined. B-stability (basis stability) of the ILP model describes set of all possible optimal solutions (Hladik 2014). Hence the optimal solution set to the ILP model will be obtained by using B-stability.

Remark 2.1 The set of constraints $A\mathbf{x} \leq \mathbf{b},\ \mathbf{x} \geq 0$, where $A \in \mathbf{A}^{\pm} \in \mathbb{IR}^{m\times n}$, $\mathbf{x} \in \mathbf{x}^{\pm} \in \mathbb{IR}^{n}$, and $\mathbf{b} \in \mathbf{b}^{\pm} \in \mathbb{IR}^{m}$, are equivalent to $A\mathbf{x} + I\mathbf{y} = \mathbf{b},\ \mathbf{x} \geq 0,\ \mathbf{y} \geq 0$, where $\mathbf{y} \in \mathbb{R}^{m}$, because there are no dependencies.

Definition 4 The model $\max\{\mathbf{c}^T\mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$ where $\mathbf{c} \in [\mathbf{c}^-, \mathbf{c}^+] \subseteq \mathbb{R}^n, A \in [A^-, A^+] \subseteq \mathbb{R}^{m\times n}$, and $\mathbf{b} \in [\mathbf{b}^-, \mathbf{b}^+] \subseteq \mathbb{R}^m$, is nominated B-stable with basis B, if B has been an optimal basis for any characteristic model. The ILP model is called (unique) non-degenerate B-stable if each version has a (unique) non-degenerate optimal basic solution within the basic B. Let $B \subseteq \{1, 2, \ldots, n\}$ be an index set so as $\mathbf{A}_B$ is non-singular and $\mathbf{A}_B$ specifies the limitation of basis indices. Similarly, $N = \{1, 2, \ldots, n\} \setminus B$ be index set for non-basic variables and $\mathbf{A}_N$ specifies the limitation of non-basic indices. B can be computed by solving a version of ILP with the centre values.

In general, B-stability conditions are as follows (Hladik 2014; Konickova 2001):

(1) *Regularity* $\mathbf{A}_B$ is regular ($\mathbf{A}_B$ is non-singular).
(2) *Feasibility* The solution set to interval system $\mathbf{A}_B\mathbf{x}_B = \mathbf{b}$ is non-negative ($\mathbf{A}_B^{-1}\mathbf{b} \geq 0$).
(3) *Optimality* $\mathbf{A}_B$ is optimal if it is feasible and $\mathbf{c}_N^T - \mathbf{c}_B^T\mathbf{A}_B^{-1}\mathbf{A}_N \leq 0^T$, when the objective function is to be maximized.

To achieve above condition we have following theorems.

Theorem 1 (Rex and Rohn 1998) *If* $\rho(|(A^c)_B^{-1}|\triangle_{A_B}) < 1$, *then* $\mathbf{A}_B$ *is regular, that* $\rho(.)$ *denotes the spectral radius.*

Theorem 2 (Rex and Rohn 1998) *If* $\max_{1\leq i\leq n}(|(A^c)_B^{-1}|\triangle_{A_B})_{ii} \geq 1$, *then* $\mathbf{A}_B$ *is not regular.*

Theorem 3 (Rohn 1993a) *If the interval vector* $\mathbf{r}$ *is an enclosure to the solution set to system* $\mathbf{A}_B\mathbf{x}_B = \mathbf{b}$, *then:*

$$r_i^- = \min\left\{-\mathbf{x}_i^* + (\mathbf{x}_i^c + |\mathbf{x}_i^c|)M_{ii}, \frac{1}{2M_{ii}-1}(-\mathbf{x}_i^* + (\mathbf{x}_i^c + |\mathbf{x}_i^c|)M_{ii})\right\},$$
$$r_i^+ = \max\left\{\mathbf{x}_i^* + (\mathbf{x}_i^c - |\mathbf{x}_i^c|)M_{ii}, \frac{1}{2M_{ii}-1}(\mathbf{x}_i^* + (\mathbf{x}_i^c - |\mathbf{x}_i^c|)M_{ii})\right\},$$

where $M = (I - |(A_B^c)^{-1}|\triangle_{A_B})^{-1}$, $\mathbf{x}^c = (A_B^c)^{-1}b^c$, $\mathbf{x}^* = M(|\mathbf{x}^c| + |(A_B^c)^{-1}|\triangle_b)$, *and* A_B^c *is non-singular and* $\rho(|(A_B^c)^{-1}|\triangle_A) < 1$.

Theorem 4 (Hladik 2014) *Let* $\mathbf{y}$ *be an interval hall of the solution set to* $A_B^T\mathbf{y} = \mathbf{c}_B$. *If* $((A_N^T)\mathbf{y})^- \geq \mathbf{c}_N^+$, *then optimality condition is ensured, when the objective function is to be maximized.*

Theorem 5 (Hladik 2014) *Let* $diag(q)$ *determines the diagonal matrix with entries* $q_1, q_2, \ldots, q_m$. *If for any* $q \in \{\pm 1\}^m$, *the solution set to system*

$$\begin{cases} ((A_B^c)^T - (\triangle_{A_B})^T diag(q))\mathbf{y} \leq \mathbf{c}_B^+, \\ -((A_B^c)^T + (\triangle_{A_B})^T diag(q))\mathbf{y} \leq -\mathbf{c}_B^-, \\ diag(q)\mathbf{y} \geq 0, \end{cases}$$

be subset of the solution set to system

$$\begin{cases} ((A_N^c)^T - (\triangle_{A_N})^T diag(q))\mathbf{y} \geq \mathbf{c}_N^+, \\ diag(q)\mathbf{y} \geq 0, \end{cases}$$

then, optimality condition is ensured.

Theorem 6 (Allahdadi et al. 2016) *Let ILP model* (1) *be unique non-degenerate B-stable, where* $B = (1, 2, \ldots, m)$. *The optimal solution set to the ILP model is equal to the intersection of the zone generated by the best model constrains and the worst model constraints with inverse sign.*

3 Review of previous solving methods of ILP

An ILP model is the union of numerous characteristic models, which are LP models. Formulating all of these characteristic models is NP-hard. In this section, we present solving methods which are previously considered for solving the ILP model. These methods solve the ILP model by formulating two sub-models. In addition, the advantages and disadvantages of each of these methods are introduced.

3.1 BWC method

In the best-worst cases (BWC) method, the ILP model (1) is transformed into two sub-models (Tong 1994):

The best sub-model:

$$\begin{aligned} \max Z^+ &= \sum_{j=1}^{n} c_j^+ x_j \\ \text{s. t.} \quad & \sum_{j=1}^{n} a_{ij}^- x_j \leq b_i^+, \quad i = 1, \ldots, m, \\ & x_j \geq 0, \quad j = 1, \ldots, n. \end{aligned} \tag{3}$$

The worst sub-model:

$$\begin{aligned} \max Z^{-} &= \sum_{j=1}^{n} c_j^{-} x_j \\ \text{s. t.} \quad & \sum_{j=1}^{n} a_{ij}^{+} x_j \leq b_i^{-}, \quad i = 1, \ldots, m, \\ & x_j \geq 0, \quad j = 1, \ldots, n. \end{aligned} \tag{4}$$

Theorem 7 (Allahdadi and Mishmast Nehi 2013; Tong 1994) *If Z^* is optimum objective function value for an arbitrary characteristic model as model* (2), *and Z^{+^*} and Z^{-^*} are optimum objective function values of model* (3) *and model* (4) *respectively, then $Z^* \in [Z^{-^*}, Z^{+^*}]$.*

Advantages (1) The BWC method obtains the best and worst optimum objective function values of the ILP model. (2) The largest and smallest feasible region of the ILP model is obtained using the constraints of the best and worst sub-models.

Disadvantages (1) A part of the solution space of the BWC method may be absolutely infeasible. (2) A part of the solution space of the BWC method may be absolutely non-optimal.

3.2 TSM method

In two-step method (TSM), only two groups of extreme constraints among the all characteristic constraints are used to transform the ILP model into two sub-models. If the objective function is to be maximized, two sub-models corresponding to Z^+ and Z^- are formulated and solved sequentially (Huang et al. 1995). Define the index sets $A_1 = \{j \mid c_j^{\pm} \geq 0\}$, and $A_2 = \{j \mid c_j^{\pm} \leq 0\}$. The first sub-model of the TSM method have been summarized as follows:

$$\begin{aligned} \max Z^{+} &= \sum_{j \in A_1} c_j^{+} x_j^{+} + \sum_{j \in A_2} c_j^{+} x_j^{-} \\ \text{s. t.} \quad & \sum_{j \in A_1} |a_{ij}^{\pm}|^{-} sign(a_{ij}^{\pm}) x_j^{+} + \sum_{j \in A_2} |a_{ij}^{\pm}|^{+} sign(a_{ij}^{\pm}) x_j^{-} \leq b_i^{+}, \quad i = 1, \ldots, m, \\ & x_j^{+} \geq 0, \quad for \;\; j \in A_1, \\ & x_j^{-} \geq 0, \quad for \;\; j \in A_2. \end{aligned} \tag{5}$$

Solving the model (5), the optimal solutions and optimum objective function value are $x_{j_{opt}}^{+}, \; (j \in A_1), \; x_{j_{opt}}^{-}, \; (j \in A_2)$, and Z_{opt}^{+}, respectively. Using the optimal solutions to model (5) the second Sub-model is expressed as follows:

$$
\begin{aligned}
\max Z^- &= \sum_{j\in A_1} c_j^- x_j^- + \sum_{j\in A_2} c_j^- x_j^+ \\
\text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^\pm|^+ sign(a_{ij}^\pm) x_j^- + \sum_{j\in A_2} |a_{ij}^\pm|^- sign(a_{ij}^\pm) x_j^+ \le b_i^-, \quad i = 1,\ldots,m, \\
& 0 \le x_j^- \le x_{j_{opt}}^+, \quad for \;\; j \in A_1, \\
& x_j^+ \ge x_{j_{opt}}^-, \qquad for \;\; j \in A_2,
\end{aligned} \tag{6}
$$

where $|a_{ij}^\pm|^- = \begin{cases} a_{ij}^-, & a_{ij}^\pm \ge 0 \\ -a_{ij}^+, & a_{ij}^\pm < 0 \end{cases}$, $|a_{ij}^\pm|^+ = \begin{cases} a_{ij}^+, & a_{ij}^\pm \ge 0 \\ -a_{ij}^-, & a_{ij}^\pm < 0 \end{cases}$, and $sign(a_{ij}^\pm) = \begin{cases} 1, & a_{ij}^\pm \ge 0 \\ -1, & a_{ij}^\pm < 0 \end{cases}$.

By solving this sub-model the optimal solutions are $x_{j_{opt}}^-, for\, j \in A_1$ and $x_{j_{opt}}^+, for\, j \in A_2$ and the optimum objective function value is Z_{opt}^-. By combining the solutions of above sub-models the results are $x_{j_{opt}}^\pm = [x_{j_{opt}}^-, x_{j_{opt}}^+], Z_{opt}^\pm = [Z_{opt}^-, Z_{opt}^+]$. In the first sub-model (i.e., model (5)), the upper bound of $b_i^\pm$ (i.e., b_i^+) is used and in the second sub-model (i.e., model (6)), the lower bound of $b_i^\pm$ (i.e., b_i^-) is used. However, in the real world applications the composition of $b_i^\pm$ and $Z^\pm$ may be different.

Advantages (1) Inexact data expressed as intervals can be inserted into the optimization process. (2) The computation of its solving process in not complicated. (3) The optimal solutions of the ILP model using this method can help to decision makers to create alternative decisions.

Disadvantages (1) The solution space of the TSM method may contain infeasible solutions (Wang and Huang 2014). (2) Some solutions of the solution space of the TSM method may be non-optimal.

3.3 MILP method

The solution space of the TSM method may contain infeasible solutions (i.e., there exist some solutions in the solution space of the TSM method which does not lie in the set $Q = \{\mathbf{x} \mid A^-\mathbf{x} \le \mathbf{b}^+, \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \ge 0\}$). To remove the infeasible solutions from the solution space of the TSM method, Zhou et al. introduced modified interval linear programming (MILP) method (Zhou et al. 2009). Define the index sets $B_1 = \{j \mid j \in A_1,\ and\ a_{ij}^\pm \ge 0\}$, $B_2 = \{j \mid j \in A_1,\ and\ a_{ij}^\pm \le 0\}$, $B_3 = \{j \mid j \in A_2,\ and\ a_{ij}^\pm \le 0\}$, and $B_4 = \{j \mid j \in A_2,\ and\ a_{ij}^\pm \ge 0\}$. The first sub-model of the MILP method is the same as the first sub-model of the TSM method (i.e., model (5). The second sub-model is formulated adding the following constraints to the second sub-model of the TSM method.

$$
\sum_{j\in B_2} -(|a_{\delta j}^\pm|^+ x_j^- - |a_{\delta j}^\pm|^- x_{j_{opt}}^+) + \sum_{j\in B_4} (|a_{\delta j}^\pm|^- x_j^+ - |a_{\delta j}^\pm|^+ x_{j_{opt}}^-) \le 0 \tag{7}
$$

where δ is the number of active constraints in model (5) as:

$$\sum_{j\in A_1} |a_{\delta j}^{\pm}|^{-} sign(a_{\delta j}^{\pm}) x_{j_{opt}}^{+} + \sum_{j\in A_2} |a_{\delta j}^{\pm}|^{+} sign(a_{\delta j}^{\pm}) x_{j_{opt}}^{-} = b_{\delta}^{+}.$$

The extra constrains (7) is formulated combining the feasibility test (i.e.,$\sum_{j=1}^{n} a_{ij}^{-} x_j \le b_i^{+}$) and active constraints of model (5). In fact, the extra constraints (7) is obtained by the following relation

$$\sum_{j=1}^{n} a_{\delta j}^{-} x_j \le b_{\delta}^{+} = \sum_{j\in A_1} |a_{\delta j}^{\pm}|^{-} sign(a_{\delta j}^{\pm}) x_{j_{opt}}^{+} + \sum_{j\in A_2} |a_{\delta j}^{\pm}|^{+} sign(a_{\delta j}^{\pm}) x_{j_{opt}}^{-}.$$

The extra constrains (7) ensure that the solution space of the MILP method is absolutely feasible.

Therefore, the second sub-model is as follows:

$$\begin{aligned}
\max Z^{-} &= \sum_{j\in A_1} c_j^{-} x_j^{-} + \sum_{j\in A_2} c_j^{-} x_j^{+} \\
\text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^{\pm}|^{+} sign(a_{ij}^{\pm}) x_j^{-} + \sum_{j\in A_2} |a_{ij}^{\pm}|^{-} sign(a_{ij}^{\pm}) x_j^{+} \le b_i^{-}, \quad i = 1,\ldots,m, \\
& 0 \le x_j^{-} \le x_{j_{opt}}^{+}, \quad for \ j \in A_1, \\
& x_j^{+} \ge x_{j_{opt}}^{-}, \qquad for \ j \in A_2, \\
& \sum_{j\in B_2} -(|a_{\delta j}^{\pm}|^{+} x_j^{-} - |a_{\delta j}^{\pm}|^{-} x_{j_{opt}}^{+}) + \sum_{j\in B_4} (|a_{\delta j}^{\pm}|^{-} x_j^{+} - |a_{\delta j}^{\pm}|^{+} x_{j_{opt}}^{-}) \le 0.
\end{aligned} \tag{8}$$

By solving sub-models (5) and (8) and combining their optimal solutions, the solution space of the MILP method is obtained.

Advantages (1) The solution space of the MILP method is absolutely feasible. (2) This method can obtain the most probable optimum objective function value.

Disadvantages (1) Some of the solutions of the solution space of the MILP method may be non-optimal. (2) The obtaining process of extra constraints is complicated.

3.4 ITSM method

The MILP method focused on how to remove the violated solutions from the solution space of the TSM method by adding extra constraint. But had not study about the reason of violated solutions. In fact, if all of the constraints $A^{-}\mathbf{x} \le \mathbf{b}^{+}$ are considered in a solving method, then its solution space pass the feasibility test. Hence, instability in selecting constraints is the key reason of violated solution. If all overlooked constraints among the constraints $A^{-}\mathbf{x} \le \mathbf{b}^{+}$ are imposed on the TSM method, then its solution space is absolutely feasible. Therefore, an improved solving method based on TSM (or ITSM) is introduced (Wang and Huang 2014). The first sub-model is the same as the first sub-model of the TSM method (i.e., model (5)) . After solving the sub-model 1, the second sub-model can be expressed as follows:

$$\begin{aligned}
&\max Z^- = \sum_{j\in A_1} c_j^- x_j^- + \sum_{j\in A_2} c_j^- x_j^+ \\
&\quad \text{s. t.} \quad \sum_{j\in A_1} |a_{ij}^\pm|^+ sign(a_{ij}^\pm) x_j^- + \sum_{j\in A_2} |a_{ij}^\pm|^- sign(a_{ij}^\pm) x_j^+ \le b_i^-, \quad i = 1,\ldots,m, \\
&\qquad \sum_{j=1}^n a_{ij}^- x_j' \le b_i^+, \quad x_j' = \begin{cases} x_j^-, & if \quad sign(a_{ij}^\pm) \ne sign(c_j^\pm), \ for \ j\in A_1, \\ x_{j_{opt}}^+, & if \quad sign(a_{ij}^\pm) = sign(c_j^\pm), \ for \ j\in A_1, \\ x_j^+, & if \quad sign(a_{ij}^\pm) \ne sign(c_j^\pm), \ for \ j\in A_2, \\ x_{j_{opt}}^-, & if \quad sign(a_{ij}^\pm) = sign(c_j^\pm), \ for \ j\in A_2, \end{cases} \\
&\qquad 0 \le x_j^- \le x_{j_{opt}}^+, \quad for \ j \in A_1, \\
&\qquad x_j^+ \ge x_{j_{opt}}^-, \qquad for \ j\in A_2.
\end{aligned} \tag{9}$$

By combining the optimal solutions of models (5) and (9), the solution space of the ITSM method can be formed. The extra constraint in the model (9) guarantee that the solution space of the ITSM method is absolutely feasible.

Advantages (1) The method to identify the extra constraints in the ITSM method is easier than ones in the MILP method. (2) The solution space of the ITSM method is absolutely feasible.

Disadvantages Some solutions obtained using the ITSM method may be non-optimal.

Note that, the extra constraints in the models (8) and (9) are equivalent. Therefore, the MILP method gets same results as the ITSM method does.

3.5 ThSM methods

Huang and Cao (2011) proposed three-step method (ThSM). If the optimal solutions obtained by the TSM method are not absolutely feasible, by constricting the solution space of the TSM method (i.e., $\mathbf{x}_{opt}^\pm = \{x_{j_{opt}}^\pm | x_{j_{opt}}^\pm = [x_{j_{opt}}^-, x_{j_{opt}}^+]\}$) into centre point, the infeasible solutions will be remove. Let $a_{ij}^\pm \ge 0$, $j = 1,\ldots,p_i$, $a_{ij}^\pm \le 0$, $j = p_i + 1, \ldots, n_0$, and $m_j = 0.5(x_{j_{opt}}^- + x_{j_{opt}}^+)$, $d_j = 0.5(x_{j_{opt}}^+ - x_{j_{opt}}^-)$. In this method, the authors want to obtain a solution space $\mathbf{y}_{opt}^\pm$ which is absolutely feasible. To do this, they constrict the solution space of the TSM method (i.e., $\mathbf{x}_{opt}^\pm = \{x_{j_{opt}}^\pm | x_{j_{opt}}^\pm = [x_{j_{opt}}^-, x_{j_{opt}}^+] = [m_j - d_j, m_j + d_j]\}$). Then, the solution space $\mathbf{y}_{opt}^\pm = \{y_{j_{opt}}^\pm | y_{j_{opt}}^\pm = [m_j - qd_j, m_j + qd_j]\}$ is defined, where $0 \le q \le 1$ is shrink rate. The solution space $\mathbf{y}_{opt}^\pm$ is absolutely feasible if $A^-\mathbf{y}_{opt}^\pm \le \mathbf{b}^\pm$. Thus

$$A^-\mathbf{y}_{opt}^\pm = \sum_{j=1}^{p_i} a_{ij}^-(m_j + qd_j) + \sum_{j=p_i+1}^{n_0} a_{ij}^-(m_j - qd_j) + \sum_{j=n_0}^{n} a_{ij}^- m_j \le b_i^+.$$

Consequently,

$$\sum_{j=1}^{p_i} a_{ij}^- q d_j - \sum_{j=p_i+1}^{n_0} a_{ij}^- q d_j \le b_i^+ - \sum_{j=1}^{n} a_{ij}^- m_j, \quad i = 1, \ldots, m.$$

By using same shrink rates for all variables in ThSM-I and different shrink rates for variables in ThSM-II, ThSM methods have been formulated as follows:

ThSM-I:

$$\begin{aligned} & \max q \\ \text{s. t. } & \sum_{j=1}^{p_i} a_{ij}^- q d_j - \sum_{j=p_i+1}^{n_0} a_{ij}^- q d_j \le b_i^+ - \sum_{j=1}^{n} a_{ij}^- m_j, \quad i = 1, \ldots, m, \\ & 0 \le q \le 1, \end{aligned} \tag{10}$$

where $m_j = 0.5(x_{j_{opt}}^- + x_{j_{opt}}^+)$, $d_j = 0.5(x_{j_{opt}}^+ - x_{j_{opt}}^-)$, and q is shrink rate, $j = 1, \ldots, p_i, p_i + 1, \ldots, n_0, n_0 + 1, \ldots, n$. Between solution $\mathbf{x}_{opt}^{\pm}$ the first n_0 decision variables are interval number, whereas the $n - n_0$ variables are crisp numbers (i.e., $q = 0$). So the final solution is as $\mathbf{y}_{opt}^{\pm} = \{y_{j_{opt}}^{\pm} | y_{j_{opt}}^{\pm} = [m_j - q d_j, m_j + q d_j]\}$.

ThSM-II:

$$\begin{aligned} & \max q_1 \times q_2 \times \cdots \times q_{n_0} \\ \text{s. t. } & \\ & \sum_{j=1}^{p_i} a_{ij}^- q_j d_j - \sum_{j=p_i+1}^{n_0} a_{ij}^- q_j d_j \le b_i^+ - \sum_{j=1}^{n} a_{ij}^- m_j, \quad i = 1, \ldots, m, \\ & 0 \le q_j \le 1, \quad j = 1, \ldots, n_0, \end{aligned} \tag{11}$$

where q_j is the shrink rate and $q_j = 0$ for last $n - n_0$ variables. The final solution can be remarked as: $\mathbf{y}_{opt}^{\pm} = \{y_{j_{opt}}^{\pm} | y_{j_{opt}}^{\pm} = [m_j - q_j d_j, m_j + q_j d_j]$.

Advantages (1) Both ThSM-I and ThSM-II are able to remove infeasible solution by constricting the solution space of the TSM method. (2) If the centre point of solution space of the TSM is feasible, then the solution spaces of this methods are absolutely feasible and optimal.

Disadvantages (1) Many feasible solutions that are near to the worst-case constraints will be eliminated. (2) The ThSM models will be infeasible when the centre point of the solution space of the TSM is not feasible. (3) The number of computational constrains will be increased, when the variables or constraints are abundant.

3.6 RTSM method

Fan and Huang (2012) used the following criteria to formulate the extra constraints.

(1) If $a_{i_o j}^{\pm} \times c_j^{\pm} \ge 0$, $\forall j$ or $a_{i_o j}^{\pm} \times c_j^{\pm} \le 0$, $\forall j$, then the solution space of the TSM method is absolutely feasible.

(2) If $a_{i_o j}^{\pm} \times c_j^{\pm} \geq 0$, for $j \in B_1 \cup B_3$ and $a_{i_o j}^{\pm} \times c_j^{\pm} \leq 0$, for $j \in B_2 \cup B_4$, then the solutions obtained using the TSM method are absolutely feasible if and only if:

$$\sum_{j \in B_1} a_{i_o j}^{-} x_j^{+} + \sum_{j \in B_2} a_{i_o j}^{-} x_{j_{opt}}^{-} + \sum_{j \in B_3} a_{i_o j}^{-} x_j^{-} + \sum_{j \in B_4} a_{i_o j}^{-} x_{j_{opt}}^{+} \leq b_{i_o}^{+}.$$

Therefore, a robust two-step method (RTSM) is introduced. In this method, the first sub-model is formulated corresponding to Z^-, and the second sub-model is formulated corresponding to Z^+, when the objective function is to be maximized. The sub-models of the RTSM method can be expressed as follows: (assume $f^{\pm} > 0, b_i^{\pm} > 0$)

Sub-model 1:

$$\begin{aligned} \max Z^- &= \sum_{j \in A_1} c_j^- x_j^- + \sum_{j \in A_2} c_j^- x_j^+ \\ \text{s. t.} \quad & \sum_{j \in A_1} |a_{ij}^{\pm}|^+ sign(a_{ij}^{\pm}) x_j^- + \sum_{j \in A_2} |a_{ij}^{\pm}|^- sign(a_{ij}^{\pm}) x_j^+ \leq b_i^-, \quad i = 1, \ldots, m, \\ & x_j^- \geq 0, \quad for \;\; j \in A_1, \\ & x_j^+ \geq 0, \quad for \;\; j \in A_2. \end{aligned} \tag{12}$$

The optimal solution to sub-model 1 are $x_{j_{opt}}^-$, $for\ j \in A_1$, and $x_{j_{opt}}^+$, $for\ j \in A_2$, and its optimum objective function value is Z_{opt}^-. Using these solutions the Sub-model 2 is as follows:

$$\begin{aligned} \max Z^+ &= \sum_{j \in A_1} c_j^+ x_j^+ + \sum_{j \in A_2} c_j^+ x_j^- \\ \text{s. t.} \quad & \sum_{j \in A_1} |a_{ij}^{\pm}|^- sign(a_{ij}^{\pm}) x_j^+ + \sum_{j \in A_2} |a_{ij}^{\pm}|^+ sign(a_{ij}^{\pm}) x_j^- \leq b_i^+, \quad i = 1, \ldots, m, \\ & \sum_{j \in B_1} a_{ij}^- x_j^+ + \sum_{j \in B_2} a_{ij}^- x_{j_{opt}}^- + \sum_{j \in B_3} a_{ij}^- x_j^- + \sum_{j \in B_4} a_{ij}^- x_{j_{opt}}^+ \leq b_i^+, \\ & x_j^+ \geq x_{j_{opt}}^-, \quad for \;\; j \in A_1, \\ & x_j^- \leq x_{j_{opt}}^+, \quad for \;\; j \in A_2, \\ & x_j^+ \geq 0, \qquad for \;\; j \in A_1, \\ & x_j^- \geq 0, \qquad for \;\; j \in A_2, \end{aligned} \tag{13}$$

where $A_j,\ \ j = 1, 2$ and $B_j,\ \ j = 1, \ldots, 4$ are defined in the Subsections 3.2 and 3.3, respectively. Therefore ultimate solutions are: $x_{j_{opt}}^{\pm} = [x_{j_{opt}}^-, x_{j_{opt}}^+], Z_{opt}^{\pm} = [Z_{opt}^-, Z_{opt}^+]$.

Advantages (1) The RTSM method may not lead to great computational needs. (2) The solution space of the RTSM method is absolutely feasible.

Disadvantage Some of the solutions obtained using the RTSM method may be non-optimal.

3.7 SOM-2 method

Lu et al. (2014) presented an alternative solution method (i.e., SOM-2) to solve the ILP model when the objective function is to be minimized (i.e., $\min f^{\pm} = \sum_{j=1}^{n} c_j^{\pm} x_j^{\pm}$), and its sub-models. In this paper we consider sub-models of the SOM-2 method when the objective function is to be maximized as model (1) (i.e., $\max f^{\pm} = \sum_{j=1}^{n} c_j^{\pm} x_j^{\pm}$). In this method, the first sub-model is formulated corresponding to Z^-, and then solved. Using the optimal solutions of the first sub-model, the second sub-model is formulated corresponding to Z^+. Compared to the TSM, this method can obtain alternative solutions which are adaptable for useful situation of decision makers. The SOM-2 method includes the following sub-models:

Sub-model 1:

$$\begin{aligned} \max Z^- &= \sum_{j\in A_1} c_j^- x_j^- + \sum_{j\in A_2} c_j^- x_j^+ \\ \text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^{\pm}|^+ sign(a_{ij}^{\pm}) x_j^- + \sum_{j\in A_2} |a_{ij}^{\pm}|^- sign(a_{ij}^{\pm}) x_j^+ \leq b_i^+, \quad i = 1,\ldots,m, \\ & x_j^- \geq 0, \quad for \;\; j \in A_1, \\ & x_j^+ \geq 0, \quad for \;\; j \in A_2. \end{aligned} \tag{14}$$

By solving above sub-model, the optimal solutions are $x_{j_{opt}}^-$, $for\; j \in A_1$, and $x_{j_{opt}}^+$, $for\; j \in A_2$, and the optimum objective function value is Z_{opt}^-. By using solutions of the first sub-model, the second sub-model can be formulated as follows:

$$\begin{aligned} \max Z^+ &= \sum_{j\in A_1} c_j^+ x_j^+ + \sum_{j\in A_2} c_j^+ x_j^- \\ \text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^{\pm}|^- sign(a_{ij}^{\pm}) x_j^+ + \sum_{j\in A_2} |a_{ij}^{\pm}|^+ sign(a_{ij}^{\pm}) x_j^- \leq b_i^-, \quad i = 1,\ldots,m, \\ & x_j^+ \geq x_{j_{opt}}^-, \qquad for \;\; j \in A_1, \\ & 0 \leq x_j^- \leq x_{j_{opt}}^+, \quad for \;\; j \in A_2. \end{aligned} \tag{15}$$

By combining the solutions of the above sub-models, final results are: $x_{j_{opt}}^{\pm} = [x_{j_{opt}}^-, x_{j_{opt}}^+], Z_{opt}^{\pm} = [Z_{opt}^-, Z_{opt}^+]$.

Advantage This method is generally simple and advantageous in some practical situation.

Disadvantages (1) In some cases the feasibility of its solution space will not be guaranteed. (2) The solution space of the SOM-2 method is not absolutely optimal in general.

In Allahdadi et al. (2016) , the authors proposed two methods to improve the MILP and TSM methods that are called improved MILP (IMILP) and improved ILP (IILP). In the following subsections, we review these methods whose solution space are absolutely both feasible and optimal.

3.8 IILP method

The solution space of the TSM method is not absolutely feasible and optimal. To obtain a solution space which is absolutely both feasibl and optimal, the IILP method is proposed to solve the ILP model. Corresponding to Z^-, the first sub-model is formulated and solved, which is the same as the second sub-model of the TSM method. In the sub-model corresponding to Z^+, an extra constraint is added to ensure that the solution space is absolutely optimal.

The first sub-model is formulated as follows:

$$\begin{aligned} \max Z^- &= \sum_{j\in A_1} c_j^- x_j^- + \sum_{j\in A_2} c_j^- x_j^+ \\ \text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^{\pm}|^+ sign(a_{ij}^{\pm}) x_j^- + \sum_{j\in A_2} |a_{ij}^{\pm}|^- sign(a_{ij}^{\pm}) x_j^+ \le b_i^-, \quad i = 1,\ldots,m, \\ & x_j^- \ge 0, \quad for \ \ j \in A_1, \\ & x_j^+ \ge 0, \quad for \ \ j \in A_2. \end{aligned} \tag{16}$$

By solving the first sub-model and using its optimal solutions, the second sub-model can be formulated as follows:

$$\begin{aligned} \max Z^+ &= \sum_{j\in A_1} c_j^+ x_j^+ + \sum_{j\in A_2} c_j^+ x_j^- \\ \text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^{\pm}|^- sign(a_{ij}^{\pm}) x_j^+ + \sum_{j\in A_2} |a_{ij}^{\pm}|^+ sign(a_{ij}^{\pm}) x_j^- \le b_i^+, \quad i = 1,\ldots,m, \\ & x_j^+ \ge x_{j_{opt}}^-, \qquad for \ \ j \in A_1, \\ & 0 \le x_j^- \le x_{j_{opt}}^+, \quad for \ \ j \in A_2, \\ & x_j^+ \ge 0, \qquad for \ \ j \in A_1, \\ & \sum_{j\in B_2} -(|a_{\delta j}^{\pm}|^- x_j^+ - |a_{\delta j}^{\pm}|^+ x_{j_{opt}}^-) + \sum_{j\in B_4} (|a_{\delta j}^{\pm}|^+ x_j^- - |a_{\delta j}^{\pm}|^- x_{j_{opt}}^+) \ge 0, \end{aligned} \tag{17}$$

where $A_j, \ j = 1,2$ and $B_j, \ j = 2,4$ are defined in the Sects. 3.2 and 3.3, respectively. The last constraint of the model (17) is an extra constraint to ensure that solution space obtained by the IILP method is absolutely optimal. In this extra constraints δ is the number of active constraints of the first sub-model at the optimal solution with its $a_{\delta j}^{\pm} \le 0, \ for \ j \in B_2 \ and \ a_{\delta j}^{\pm} \ge 0, \ for \ j \in B_4$. After solving above sub-models the result is $x_{j_{opt}}^{\pm} = [x_{j_{opt}}^-, x_{j_{opt}}^+], Z_{opt}^{\pm} = [Z_{opt}^-, Z_{opt}^+]$.

Advantage The solution space of the IILP method is absolutely both feasible and optimal.

3.9 IMILP method

The solution space of the MILP method is absolutely feasible, but is not absolutely optimal. To remove the non-optimal solutions from the solution space of the MILP method, an improved MILP (IMILP) method is introduced (Allahdadi et al. 2016).

In the IMILP method the ILP model is transformed into two sub-models. The first sub-model of the IMILP method is the same as the first sub-model of the MILP method. In Allahdadi et al. (2016) the authors proved that, sign of the main constraints of the second sub-model of the MILP should be reversed, because by reversing the sign of inequalities in the main constrains of the second sub-model of the MILP method, optimality of the feasible solution space will be ensured. The second sub-model can be expressed as follows:

$$\begin{aligned} \min Z^- &= \sum_{j\in A_1} c_j^- x_j^- + \sum_{j\in A_2} c_j^- x_j^+ \\ \text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^\pm|^+ sign(a_{ij}^\pm) x_j^- + \sum_{j\in A_2} |a_{ij}^\pm|^- sign(a_{ij}^\pm) x_j^+ \geq b_i^-, \quad i = 1, \ldots, m, \\ & 0 \leq x_j^- \leq x_{j_{opt}}^+, \quad for \;\; j \in A_1, \\ & x_j^+ \geq x_{j_{opt}}^-, \qquad for \;\; j \in A_2, \\ & \sum_{j\in B_2} -(|a_{\delta j}^\pm|^+ x_j^- - |a_{\delta j}^\pm|^- x_{j_{opt}}^+) + \sum_{j\in B_4} (|a_{\delta j}^\pm|^- x_j^+ - |a_{\delta j}^\pm|^+ x_{j_{opt}}^-) \leq 0, \end{aligned} \tag{18}$$

where $A_j, \;\; j = 1, 2$ and $B_j, \;\; j = 2, 4$ are defined in the Sects. 3.2 and 3.3, respectively. Note that the objective function is changed from maximize to minimize, because the sign of the main constraints are reversed. The last constraint of model (18) is the same as the extra constrains of the MILP method. This extra constrains ensures the feasibility of solutions. By solving and combining solutions of sub-models of the IMILP method we have $x_{j_{opt}}^\pm = [x_{j_{opt}}^-, x_{j_{opt}}^+], Z_{opt}^\pm = [Z_{opt}^-, Z_{opt}^+]$.

Advantage The solution space of this method is absolutely both feasible and optimal.

4 Improved method

In the SOM-2 method two sub-models were considered to solve the ILP model (1). Note that, each of the sub-models of the SOM-2 method is only an arbitrary characteristic model of the ILP model (1), and the solution space of the SOM-2 method is not absolutely feasible and optimal in general. In this section we try to improve the SOM-2 method such that the solution space is both feasible and optimal. The improved SOM-2 method is called ISOM-2 method.

Theorem 8 (Zhou et al. 2009) *To Describe the relationships between decision variables* $\mathbf{x}^\pm$ *and left-hand sides (constraint coefficients) of the ILP model* (1), *the coefficients of* x_j^- *should be* $|a_{ij}^\pm|^+ sign(a_{ij}^\pm)$ *for* $j \in A_1$, *and the coefficients of* x_j^+ *should be* $|a_{ij}^\pm|^- sign(a_{ij}^\pm)$ *for* $j \in A_2$, *for the first sub-model* Z^-, *and the coefficients of* x_j^+ *should be* $|a_{ij}^\pm|^- sign(a_{ij}^\pm)$ *for* $j \in A_1$, *and the coefficients of* x_j^- *should be* $|a_{ij}^\pm|^+ sign(a_{ij}^\pm)$ *for* $j \in A_2$, *for the second sub-model* Z^+.

Theorem 9 (Zhou et al. 2009) *To ensure that solution space* $\mathbf{x}^{\pm}$ *is absolutely feasible, the following constraints should be added to constraints of second sub-model*

$$\sum_{j\in B_1}(|a^{\pm}_{\delta j}|^{-}x^{+}_{j} - |a^{\pm}_{\delta j}|^{+}x^{-}_{j_{opt}}) + \sum_{j\in B_3} -(|a^{\pm}_{\delta j}|^{+}x^{-}_{j} - |a^{\pm}_{\delta j}|^{-}x^{+}_{j_{opt}}) \leq 0, \tag{19}$$

where δ *is the number of active constraints in the first sub-model and* $B_j,\ \ j = 1, 3$ *are defined in the Sect.* 3.3.

By checking effect of the extra constraint (19) $\forall\delta$, on feasible region of the second sub-model of the SOM-2 method, some of them may be redundant. Define the index sets D_1 and D_2 as follows:

$$D_1 = \{i \mid i\text{th } \textit{extra constraint through the constraints } (19) \textit{ is redundant } \},$$
$$D_2 = \{i \mid i\text{th } \textit{extra constraint through the constraints } (19) \textit{ is not redundant } \}.$$

Theorem 10 *Let the ILP model* (1) *be unique non-degenerate B-stable with* $B = (1, \ldots, m)$. *To ensure that solution space* $\mathbf{x}^{\pm}$ *is optimal, sign of inequalities of some of the main constraints of the second sub-model of the SOM-2 method, that their indices belong to* D_2, *should be reversed as follows:*

$$\sum_{j\in A_1}|a^{\pm}_{\delta j}|^{-}sign(a^{\pm}_{\delta j})x^{+}_{j} + \sum_{j\in A_2}|a^{\pm}_{\delta j}|^{+}sign(a^{\pm}_{\delta j})x^{-}_{j} \geq b^{-}_{\delta}, \tag{20}$$

where $\delta \in D_2$.

Proof Similar to proof of Theorem 4.5 from Allahdadi et al. (2016), this theorem is proved. □

Not that, in some other main constraints of the second sub-model of the SOM-2 method, that their indices belong to D_1 sign of inequalities should not be changed, and are as follows:

$$\sum_{j\in A_1}|a^{\pm}_{\delta j}|^{-}sign(a^{\pm}_{\delta j})x^{+}_{j} + \sum_{j\in A_2}|a^{\pm}_{\delta j}|^{+}sign(a^{\pm}_{\delta j})x^{-}_{j} \leq b^{-}_{\delta}, \tag{21}$$

where $\delta \in D_1$.

Thus, in ISOM-2 method two sub-models can be formulated for solving the ILP model (1). The first sub-model is the same as the first sub-model of the SOM-2 method as follows:

$$\begin{aligned} \max Z^{-} &= \sum_{j\in A_1} c^{-}_{j}x^{-}_{j} + \sum_{j\in A_2} c^{-}_{j}x^{+}_{j} \\ \text{s. t.} \quad & \sum_{j\in A_1}|a^{\pm}_{ij}|^{+}sign(a^{\pm}_{ij})x^{-}_{j} + \sum_{j\in A_2}|a^{\pm}_{ij}|^{-}sign(a^{\pm}_{ij})x^{+}_{j} \leq b^{+}_{i}, \quad i = 1, \ldots, m, \\ & x^{-}_{j} \geq 0, \quad for \ \ j \in A_1, \\ & x^{+}_{j} \geq 0, \quad for \ \ j \in A_2. \end{aligned} \tag{22}$$

By solving model (22) and using its optimal solutions and by attention to Theorems 9 and 10 the second sub-model will be formulated. By attention to Theorem 9, to ensure that solutions are absolutely feasible we add the extra constraints (19) to the second sub-model. By considering extra constraints (19) for all active constraints of the first sub-model, some of these extra constraints in the second sub-model of SOM-2 are redundant, and we can remove these redundant constraints through the second sub-model of the SOM-2 method. For some other constraints, the corresponding extra constraints are not redundant and may change the optimal solution of model. In some of the main constrains of the second sub-model of the SOM-2 method that their corresponding extra constrains are not redundant, the sign of inequalities are reversed by attention to Theorem 10. Note that, in other main constrains of the second sub-model, in which the corresponding extra constrains are redundant, sign of inequalities are not changed. Hence, by Theorem 9 feasibility of solutions will be ensured by adding the extra constraints (19), and by Theorem 10 the constraints (20) guarantee that solutions of this method are optimal. So, the second sub-model of ISOM-2 method can be summarized as follows:

$$\begin{aligned}
\max Z^{+} &= \sum_{j\in A_1} c_j^{+}x_j^{+} + \sum_{j\in A_2} c_j^{+}x_j^{-} \\
\text{s. t.} \quad & \sum_{j\in A_1} |a_{ij}^{\pm}|^{-} sign(a_{ij}^{\pm})x_j^{+} + \sum_{j\in A_2} |a_{ij}^{\pm}|^{+} sign(a_{ij}^{\pm})x_j^{-} \geq b_i^{-}, \quad for \;\; i\in D_2, \\
& \sum_{j\in A_1} |a_{ij}^{\pm}|^{-} sign(a_{ij}^{\pm})x_j^{+} + \sum_{j\in A_2} |a_{ij}^{\pm}|^{+} sign(a_{ij}^{\pm})x_j^{-} \leq b_i^{-}, \quad for \;\; i\in D_1, \\
& x_j^{+} \geq x_{j_{opt}}^{-}, \qquad for \;\; j\in A_1, \\
& 0 \leq x_j^{-} \leq x_{j_{opt}}^{+}, \quad for \;\; j\in A_2, \\
& \sum_{j\in B_1} (|a_{\delta j}^{\pm}|^{-}x_j^{+} - |a_{\delta j}^{\pm}|^{+}x_{j_{opt}}^{-}) + \sum_{j\in B_3} -(|a_{\delta j}^{\pm}|^{+}x_j^{-} - |a_{\delta j}^{\pm}|^{-}x_{j_{opt}}^{+}) \leq 0.
\end{aligned} \tag{23}$$

The last constraint of model (23) is an extra constraint to ensure that solution space obtained using the ISOM-2 method is absolutely feasible, where δ is the number of active constraints of the first sub-model. The first constraint of model (23) is considered to ensure that the solution space is absolutely optimal. The sets $A_j, \;\; j=1,2$, $B_j, \;\; j=1,3$ and $D_i \;\; i=1,2$ are defined in the Sects. 3.2, 3.3, and 4, respectively.

By combining the optimal solutions of the above sub-models, final results are: $x_{j_{opt}}^{\pm} = [x_{j_{opt}}^{-}, x_{j_{opt}}^{+}], Z_{opt}^{\pm} = [Z_{opt}^{-}, Z_{opt}^{+}]$.

Corollary 1 *By assumption of basis stability of the ILP model, if* $\mathbf{x}_{opt}^{\pm}$ *is solution space of the ISOM-2 method, then* $\mathbf{x}_{opt}^{\pm}$ *is both absolutely feasible and optimal.*

Proof Let $\mathbf{x}_{opt}^{\pm}$ is the solution space of the ISOM-2 mthod.

If $\sum_{j=1}^{n} a_{ij}^{-}x_j^{'} = max\{\sum_{j=1}^{n} a_{ij}^{-}x_j \mid x_j \in x_j^{\pm}\} \leq b_i^{+}$, then $\mathbf{x}_{opt}^{\pm}$ is absolutely feasible. According to index sets $B_i, i=1,\ldots,4$ denoted in Sect. 3.3, we have $a_{ij}^{-} \geq 0, j\in B_1 \bigcup B_4$ and $a_{ij}^{-} \leq 0, j \in B_2 \bigcup B_3$. Since $x_j \geq 0, \forall j$, we have $a_{ij}^{-}x_j \leq a_{ij}^{-}x_{j_{opt}}^{+}, \forall j \in B_1 \bigcup B_4$, and $a_{ij}^{-}x_j \leq a_{ij}^{-}x_{j_{opt}}^{-}, \forall j \in B_2 \bigcup B_3$. Thus,

$$(A^-)_i\mathbf{x} = a_{i1}^- x_1 + \cdots + a_{in}^- x_n \leq \sum_{j\in B_1\bigcup B_4} a_{ij}^- x_{j_{opt}}^+ + \sum_{j\in B_2\bigcup B_3} a_{ij}^- x_{j_{opt}}^-, i = 1,\ldots,m.$$

Then, $\mathbf{x}' = [x_{j_{opt}}^+, \forall j \in B_1 \bigcup B_4, x_{j_{opt}}^-, \forall j \in B_2 \bigcup B_3]$. The extreme point $\mathbf{x}'$ is a vertex at the polyhedron of the solution space $\mathbf{x}_{opt}^{\pm}$. By Theorem 9, Eq. (19) would ensure that the solution space $\mathbf{x}_{opt}^{\pm}$ is absolutely feasible. Similarly, if $\sum_{j=1}^n a_{ij}{}^+ x_j{}'' = min\{\sum_{j=1}^n a_{ij}{}^+ x_j \mid x_j \in x_j^{\pm}\} \geq b_i{}^-$, then the solution space $\mathbf{x}_{opt}^{\pm}$ is absolutely optimal. Based on the same principle $\mathbf{x}'' = [x_{j_{opt}}^-, \forall j \in B_1 \bigcup B_4, x_{j_{opt}}^+, \forall j \in B_2 \bigcup B_3]$ is a vertex at the polyhedron of the solution space $\mathbf{x}_{opt}^{\pm}$. By Theorem 10, Eq. (20) would ensure that the solution space $\mathbf{x}_{opt}^{\pm}$ is absolutely optimal. Hence, the solution space of the ISOM-2 method is both absolutely feasible and optimal. □

Note that, the interval of objective function of ISOM-2 method is subset of interval of exact objective function of ILP model (1), that is obtained by the BWC method.

Advantage The solution space of the ISOM-2 method is absolutely feasible and optimal.

5 Numerical example

In this paper we consider an example to obtain the solution space of solving methods of the ILP model (1). consider the following ILP model:

$$\begin{aligned} &\max \quad Z = [3, 3.5]x_1 - [1, 1.2]x_2 \\ &\text{s. t.; } [1, 1.1]x_1 + [1.6, 1.8]x_2 \leq [11.6, 12] \\ &\quad [3, 4]x_1 - [2, 3]x_2 \leq [5, 7] \\ &\quad x_1, x_2 \geq 0. \end{aligned} \tag{24}$$

The feasible region of the worst and best sub-models of the ILP model (24) are shown in Fig. 1. These regions are the smallest and largest feasible region of the feasible region of the ILP model (24), respectively. In addition, the extreme zones of the feasible region of the ILP model (24) are shown in Fig. 1. Note that, the region between the feasible region of the worst model and the feasible region of the best model is the ridge of the feasible region of the ILP model which includes the extreme zones. Obviously, the unique optimal solution to a characteristic model lies in the set of extreme point of its feasible region. Because the ILP model is the union of its characteristic models, the unique optimal solution set to the ILP model is one of the extreme zones which are shown in Fig. 1. Note that, all points of the feasible region of the best model are also feasible solutions of the ILP model. Among the feasible solutions some of them, that lie in the ridge of the feasible region of the ILP model, may be optimal. If the ILP model is B-stable, the optimal solutions only lie in one of the extreme zones.

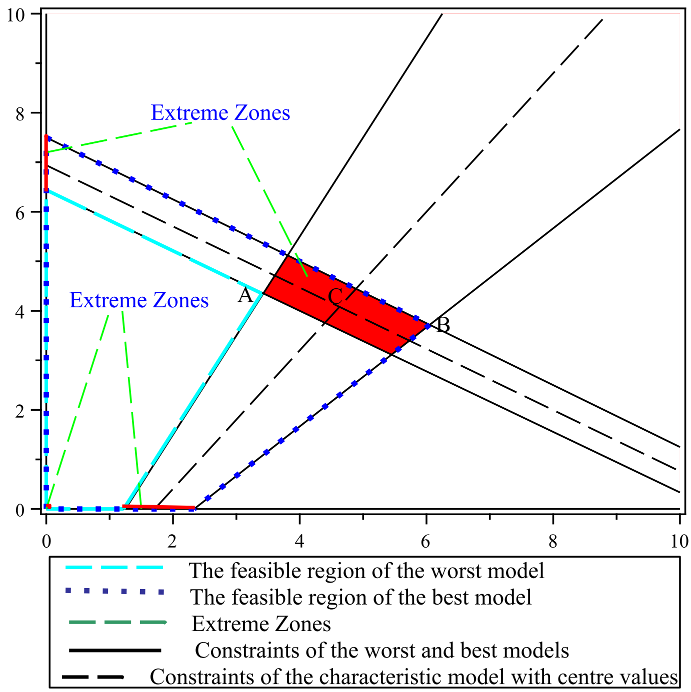

Fig. 1 The feasible region and extreme zones of the ILP model

By Remark 2.1 the standard form of the ILP model (24) is as follows:

$$\begin{aligned} &\max \quad Z = [3, 3.5]x_1 - [1, 1.2]x_2 \\ &\text{s. t.}[1, 1.1]x_1 + [1.6, 1.8]x_2 + x_3 = [11.6, 12] \\ &\quad [3, 4]x_1 - [2, 3]x_2 + x_4 = [5, 7] \\ &\quad x_1, x_2, x_3, x_4 \geq 0. \end{aligned} \tag{25}$$

By solving characteristic model with centre values, $x_1, x_2 > 0 and x_3, x_3 = 0$. Hence, $\mathbf{x}_B = (x_1, x_2)$ and so a candidate basis for B-stability is $B = (1, 2)$. The constraints of the characteristic model with centre values and its optimal solution (i.e., C(4.63,4.08)) are shown in Fig. 1.

1. Unto Theorem 1, $\mathbf{A}_B = \begin{pmatrix} [1, 1.1] & [1.6, 1.8] \\ [3, 4] & [-3, -2] \end{pmatrix}$ is regular because the spectral radius is equal to 0.21.
2. Using Theorem 3, we can calculate the enclosure to the solution set to $\mathbf{A}_B\mathbf{x}_B = \mathbf{b}$, to be $\mathbf{x}_B = \begin{pmatrix} x_1 = [3.34, 6.29] \\ x_2 = [3.07, 5.34] \end{pmatrix}$ because $\mathbf{x}_B^- \geq 0$, then the feasibility is valid.
3. Reusing Theorem 3, we can calculate the enclosure to the solution set to $A_B^T\mathbf{y} = \mathbf{c}_B$, to be $\mathbf{y} = \begin{pmatrix} y_1 = [0.03, 1.03] \\ y_2 = [0.6, 1.03] \end{pmatrix}$.

According to Theorem 4, because $((A_N^T)\mathbf{y})^- \geq \mathbf{c}_N^+$, then the optimality condition is valid. Therefore the ILP model is B-stable.

Because the ILP model (24) is B-stable, its exact optimal solution set is equal to solution set to the system $\mathbf{A}_B^- \mathbf{x}_B \leq \mathbf{b}^+, \mathbf{A}_B^+ \mathbf{x}_B \geq \mathbf{b}^-, \mathbf{x}_B \geq \mathbf{0}, \mathbf{x}_N = \mathbf{0}$. Therefore, the exact optimal solution set to the ILP model (24) is equal to

$$\begin{aligned} U = &\{(x_1, x_2) : x_1 + 1.6x_2 \leq 12,\ 3x_1 - 3x_2 \leq 7,\ 1.1x_1 + 1.8x_2 \geq 11.6, \\ &4x_1 - 2x_2 \geq 5,\ x_1, x_2 \geq 0\}. \end{aligned}$$

The set U is an extreme zone of the feasible region of the ILP model (24), which is shown in Fig. 2.

Note that, the optimal solutions of the worst and best models are denoted by points A and B in Figures, respectively. Therefore, the set of all solutions (such as A, B, C, et cetera) that are optimal for a given characteristic model is the exact optimal solution set to the ILP model (24).

Now, we solve the ILP model (24) using the BWC, TSM, MILP, SOM-2, ITSM, ThSM-I, ThSM-II, and RTSM methods. The solution spaces and optimum objective function values obtained using these methods are summarized in Table 1.

The exact interval of optimum objective function values is obtained using the BWC method and is [5.06, 17.46]. Because of adding extra constraint to the second sub-model of the MILP and ITSM methods to ensure absolute feasibility of their solution space, the terminal result Z^- of these methods is 4.91, that is less than the terminal result of the BWC method (i.e., 5.06). This mean that the solution spaces of the MILP and ITSM methods include some solutions whose optimum objective functions does not lie in [5.06, 17.46]. For example, the point $(x_1, x_2) = (3.19, 3.88)$ lies in the solution spaces of the MILP and ITSM methods, while its corresponding optimum objective function is 4.91 which does not lie in [5.06, 17.46]. In other point of view, the point $(x_1, x_2) = (3.19, 3.88)$ does not satisfy the optimality constraint $1.1x_1 + 1.8x_2 \geq 11.6$, because $1.1 \times 3.19 + 1.8 \times 3.88 = 10.49 \ngeq 11.6$.

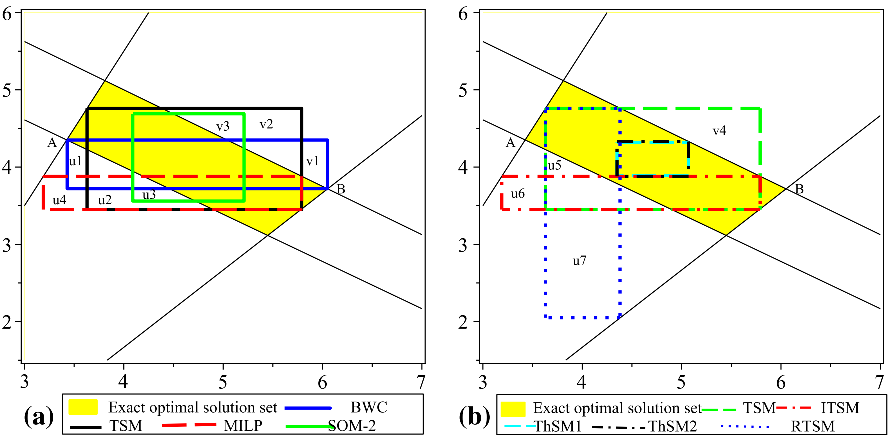

Fig. 2 Exact optimal solution set to the ILP model (24), **a** The solution spaces of the BWC, TSM, MILP, and SOM-2 methods, and **b** The solution spaces of the TSM, ITSM, ThSM-I, ThSM-II, and RTSM methods

Table 1 Results obtained using the solving methods of the ILP model (24)

Methods	$x^{\pm}_{1_{opt}}$	$x^{\pm}_{2_{opt}}$	$Z^{\pm}_{opt}$
BWC	[3.43,6.05]	[3.72,4.35]	[5.06,17.46]
TSM	[3.63,5.79]	[3.45,4.76]	[5.18,16.80]
MILP	[3.19,5.79]	[3.45,3.88]	[4.91,16.80]
$SOM-2$	[4.09,5.21]	[3.56,4.69]	[6.66,14.75]
ITSM	[3.19,5.79]	[3.45,3.88]	[4.91,16.80]
ThSM(I)	[4.35,5.07]	[3.89,4.32]	[7.86,13.86]
ThSM(II)	[4.35,5.07]	[3.88,4.33]	[7.87,13.85]
RTSM	[3.63,4.38]	[2.05,4.76]	[5.18,13.29]

Therefore, this point is not an optimal solution to the ILP model (24), while it is a feasible point of the feasible region of the ILP model. Thus, the solution spaces of the MILP and ITSM methods include non-optimal solutions, however these are absolutely feasible. Similarly, the solution spaces of other methods can be analyzed.

In Figs. 2 and 3, the exact optima solution set to the ILP model (24) denoted by yellow region, which is equal to the intersection of the region generated by the best model constraints and the worst model constraints with inverse sign, since the ILP model is B-stable. In this Figures we compare the solution spaces of the solving methods with the exact optimal solution set to the ILP model. In Fig. 2a, the exact optimal solution set and the solution space of the BWC, TSM, MILP, and SOM-2 methods are shown. Note that, triangle zones v1, v2 and v3 are not feasible and zones u1, u2, u3 and u4 are not optimal. The solution spaces of the BWC, TSM, and SOM-2 methods includes infeasible parts v1, v2 and v3 respectively and the solution spaces of these methods contains parts u1, u2 and u3 that are not optimal. However the solution space of the MILP method is absolutely feasible, part u4 of the solution space of the MILP method is not optimal.

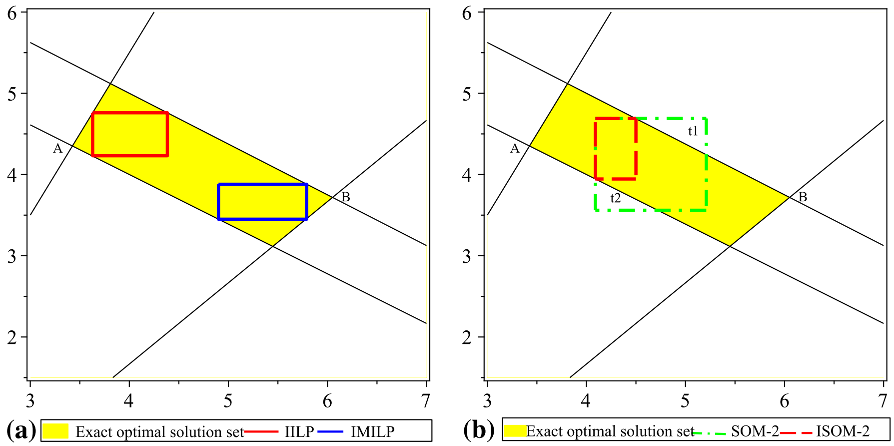

Fig. 3 Exact optimal solution set to the ILP model (24), **a** The solution spaces of the IILP and IMILP methods, and **b** The solution spaces of the SOM-2 and ISOM-2 methods

Similarly, the solution spaces of the TSM, ITSM, ThSM-I, ThSM-II, and RTSM are shown in Fig. 2b. The triangle part v4 of the solution space of the TSM method is not feasible and the triangle part u5 is not optimal. The solution spaces of the ITSM and RTSM methods is absolutely feasible but trapezoidal parts u6 and u7 of the solution spaces of these methods are not optimal. However the solution space of ThSMs method in this example is absolutely feasible and optimal, but it is not general. In other words, both ThSM-I and ThSM-II are able to eliminate infeasible solutions by shrinking the solution space of the TSM method into its centre point. Additionally, if the centre point is infeasible, both ThSM-I and ThSM-II become infeasible. Accordingly, in the solution spaces obtained using the BWC, TSM, and SOM-2 methods, part of the solution space is absolutely infeasible and part of the solution space is not optimal. In the MILP, ITSM, and RTSM methods, adding extra constraint to the sub-model 2 ensures that their solution spaces are absolutely feasible, while the solution spaces of these methods include non-optimal solutions. The ThSMs method will be infeasible if centre point of the solution space of the TSM method is infeasible. Therefore, the above methods do not study the absolute optimality of their solution spaces.

Now, we solve the ILP model (24) using the IILP, IMILP, and ISOM-2 methods. The result summarized in Table 2 and Fig. 3.

The solution spaces of the IILP and IMILP methods are shown in Fig. 3a. The solution spaces of these methods are absolutely feasible and optimal. In the Fig. 3b, the solution spaces of the SOM-2 and ISOM-2 methods are shown. Part t1 of the solution space of the SOM-2 method is not feasible and part t2 is not optimal. Obviously, the solution space of the ISOM-2 method is both absolutely feasible and optimal, while the solution space of the SOM-2 method is not.

The range of optimum objective function values of the ILP model (24) using the solving methods are shown in Fig. 4. The exact interval of the optimum objective function values is obtained using the BWC method. The interval of optimum objective function values obtained using the MILP and ITSM methods is not subset of the exact interval of the optimum objective function values. The intervals of optimum objective function values obtained using the other methods are subset of the exact interval of the optimum objective function values. Note that, a given point is an optimal solution of the ILP model if it lies in the exact optimal solution set of the ILP model. For example, consider the feasible point $(x_1, x_2) = (3.43, 3.72)$. The interval of objective function of the ILP model at this point is $[Z^-, Z^+] = [5.83, 8.28]$ which is subset of [5.06, 17.46], but it is not a reason that this point be optimal. Because this point does not satisfy the optimality constraint

Table 2 Results obtained using the IILP, IMILP and ISOM-2 methods for solving the ILP model (24)

Methods	$x^{\pm}_{1_{opt}}$	$x^{\pm}_{2_{opt}}$	$Z^{\pm}_{opt}$
IILP	[3.63,4.384]	[4.231,4.76]	[5.18,11.113]
IMILP	[4.9,5.79]	[3.45,3.88]	[10.04,16.80]
$ISOM-2$	[4.09,4.5]	[3.945,4.69]	[6.66,11.80]

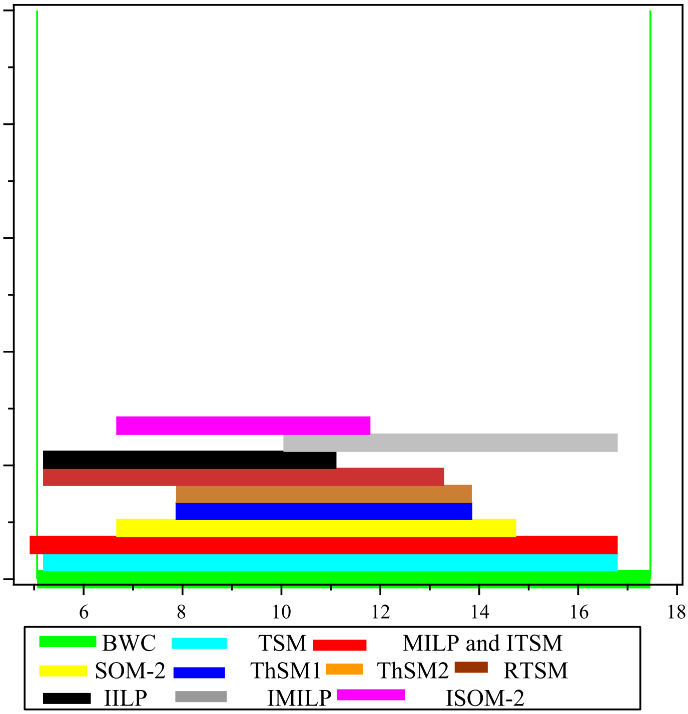

Fig. 4 The range of optimum objective function values of the ILP model (24) using the solving methods

$1.1x_1 + 1.8x_2 \geq 11.6$. Therefore, two conditions feasibility and optimality of solutions must be considered together.

Therefore, among the above-mentioned solving methods, the IILP, IMILP and ISOM-2 methods are applicable, since their solution space are both absolutely feasible and optimal and their optimum objective function values are subset of [5.06, 17.46].

6 Conclusion

In this paper, we first reviewed some previous methods for solving ILP models from the feasibility and optimality points of view. These methods transform the ILP model into two sub-models. Compared to the exact optimal solution set of the ILP model, the solution spaces of some methods include infeasible solutions, such as BWC, TSM, and SOM-2. Also, the solution space obtained by some methods contain non-optimal solutions, such as ITSM, MILP, and ThSM. Some other methods, such as IILP and IMILP, obtain a solution space under the assumption of basis stability. These methods guarantee that the solutions are absolutely feasible and optimal. In the SOM-2 method, both feasibility and optimality are not established, in general. Therefore, an improved method namely ISOM-2 is introduced to delete infeasible and non-optimal solutions in the solution space of the SOM-2 method.

In future, we will introduce new methods or extension of above-mentioned methods to obtain further optimal solutions of the ILP model.

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